

Osman Can Kandemiroglu

BIOGEOSCIENTIST — ORGANIC-MATTER TRANSFORMATION & TRACE-ELEMENT MOBILISATION

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Works across microbial ecology, organic geochemistry and aqueous trace-element chemistry: quantitative incubation and degradation experiments with living bacteria, ^{13}C stable-isotope tracing, extraction-free molecular imaging, and the Python analysis behind the measurements. In hydrogeochemistry at Bochum (2010–2011, 2016–2018): where a trace metal is released from a solid phase, what drives the release, and what takes it back out of solution — a master's thesis and peer-reviewed paper on nickel, and a doctoral project on trace elements and trace organic substances in groundwater and drinking-water treatment. Category B driving licence.

EDUCATION

M.Sc. Marine Geosciences

04/2020 – 03/2026

University of Bremen · awarded 23.03.2026, good (2.32), 120 CP, English

Thesis (30 CP): *Instant Visualization of Organic Matter Transformation in the Ocean* — MARUM, BIOM & MPI for Marine Microbiology, within DFG TRR 420 CONCENTRATE. Examiners: Prof. Dr. K.-U. Hinrichs, Prof. Dr. J.-H. Hehemann. Modules incl. Environmental Archives Projects (1.0), Research Seminar (1.7).

M.Sc. Geosciences

2008 – 2011

Ruhr-Universität Bochum · good (70/100); thesis very good (80/100)

Thesis: origin of nickel in raw water of the Breyell & Kaldenkirchen waterworks and its hydrochemical behaviour during treatment. Published, peer-reviewed, 2012.

Doctoral study, Geosciences

2016 – 2018

Ruhr-Universität Bochum

Behaviour of trace elements and trace organic substances in groundwater and drinking-water treatment. Supervisors Prof. Dr. F. Wisotzky, Prof. Dr. S. Wohnlich. Discontinued when the first supervisor fell ill and took early retirement.

Guest student, Geochemistry

2019 – 2021

Utrecht University

Aquatic chemistry · marine microbes & biogeochemistry · organic geochemistry · physical chemistry.

B.Sc. Geophysical Engineering

2002 – 2006

Kocaeli Üniversitesi · 240 ECTS

RESEARCH EXPERIENCE

Guest Scientist, AG Hehemann — BIOM / MPI Marine Microbiology, Bremen

10/2024 – 12/2025

Research Student, MPI Bremen, 05/2025–12/2025

- Five particle-degradation experiments (ADE I–V) on seven marine *Vibrio* strains, mono- and co-culture; >1,600 particles tracked by time-lapse microscopy on a custom calcium-alginate microparticle substrate and defined minimal-medium family (M0–M4).
- Separated two colonisation strategies — $78.1 \pm 17.6\%$ vs $50.5 \pm 24.2\%$ particle-volume loss ($P=1.7 \times 10^{-10}$) — and identified phosphate as the nutrient switch governing the trade-off.
- Converted single-particle measurements to fluxes: ImageJ sizing → degradation constant → Martin-curve export exponent b $0.78 - 0.95$; 96-well growth kinetics, Hungarian-algorithm tracking; all analysis in Python 3.11.

Researcher, MARUM, AG Hinrichs

08/2023 – 09/2024

Light-driven structuring of a hypersaline microbial mat (Mallorca)

- Developed extraction-free MALDI-FT-ICR-MS imaging (Bruker solarix XR 7T, $50\mu\text{m}$) for intact biomarkers measured on the particle itself, with no extraction step to bias it.
- Resolved eleven biomarker-defined redox zones across two mats; discovered PC-DEG C44:3 and 14 associated new sulfate-reducing-bacteria biomarkers.
- Integrated molecular imaging with μ -XRF elemental mapping (S, Fe, Mn, Br) and microsensor profiles; correlation-network analysis in Cytoscape.

Student Assistant — AWI Organic Geochemistry Labs, Bremerhaven

05/2022 – 01/2023

Supervisor Dr. Juliane Müller

- Full lipid-biomarker pipeline on IODP Expedition 374 (Ross Sea), PS81/PS119: extraction, open-column LC, silylation, then identification/quantification of highly branched isoprenoids and GDGTs by GC-MS and HPLC-MS; operated GC-FID and HPLC-APCI-MS.

^{13}C Stable-Isotope Probing — Helgoland Mud Area

2022

MARUM, North Sea

- Three-arm ^{13}C -labelled incubations (48h/28°C, 120h/28°C, 120h/8°C) vs a frozen t_0 control, four sediment depths (23–380cm bsf), Heincke HE595 core.
- Traced label into fatty acids by GC-C-IRMS; bacteria responded to warming faster than archaea; resolved an ANME-1 → ANME-2 transition at ~175cm from the OH-archaeol/archaeol ratio.

Doctoral Researcher & TA, Hydrogeochemistry — RUB

2016 – 2018

Trace elements & trace organics in groundwater and water treatment

- PHREEQC/AquaChem speciation and mineral-saturation modelling of a redox-stratified aquifer (As, Pb, Cd, Cr, Ni, Cu, Zn, Se, Sb, Al) under parallel Fe(II)/Fe(III) and measured-pe treatments.
- Fellerhöfe deep-well project (Wasserwerke Willich): project coordination; designed interval-resolved sampling (120 samples, 0–185m) for four production wells; delivered client presentation, Willich, 19.02.2018.
- Organic trace substances (pharmaceuticals, pesticides) in NRW aquifers — conference paper, 2018. Four lecture courses over four semesters; RUB Research School TA (RS PLUS, WiSe 17/18).

Nickel Mobilisation & Fixation — Stadtwerke Nettetal / RUB

2010 – 2011

M.Sc. thesis; peer-reviewed, gwf Wasser|Abwasser, 2012

- ~40,000 data points across 1,506 groundwater samples (73 wells, 1983–2009), 144 borehole profiles condensed into 5 hydrogeological profiles, 28 filter-material samples.
- Release driven by acidification below pH 6.5; fixation onto manganese oxides ($r^2=0.66$) and calcium-carbonate minerals ($r^2=0.86$) in the filter bed, vs the $20\mu\text{g/L}$ drinking-water limit.
- Selective extraction (Na-acetate) and aqua regia digestion per DIN EN 13346; field collaboration with Stadtwerke Nettetal.

- Multi-proxy limnic–marine transition reconstruction at 8.5–8.0 ka: GDGT/alkenone biomarkers (TEX₈₆H, BIT, U^K₃₇, brGDGT-MAT) with BAYSPAR, XRF core scanning, palynology, environmental magnetics. Manuscript in internal revision.

METHODS

Experimental design & cultivation — incubation/degradation experiments · long-term & anaerobic incubations · ¹³C stable-isotope probing · marine bacterial mono-/co-culture · growth kinetics (OD₆₀₀, 96-well) · defined minimal-medium design · microparticle synthesis · time-lapse microscopy & quantitative image analysis (ImageJ, EVOS FL Auto).

Molecular & spectroscopic — MALDI/LDI-FT-ICR-MS imaging (Bruker solariX XR 7T) · GC-MS · HPLC-MS (ESI/APCI) · GC-C-IRMS · GC-FID · lipid extraction, open-column LC, silylation, FAME prep · IPLs, GDGTs, HBIs, alkenones, pigments, quinones · μ-XRF · microsensor profiling (O₂, H₂S, pH, redox) · Bruker FlexImaging/DataAnalysis, Xcalibur, Isodat.

PUBLICATIONS & CONFERENCE CONTRIBUTIONS

Peer-reviewed journal article

Wisotzky, F., **Kandemiroglu, O.C.** & Plassmann, C. (2012). Nickelfreisetzung in das Grundwasser und dessen Bindung bei der Wasseraufbereitung zu Trinkwasser (Nettetal/Niederrhein). *gwf Wasser | Abwasser*, 153(7), 828–832.

Conference contributions & posters

Kandemiroglu, O.C., Gu, B. & Hehemann, J.-H. (2025). Instant Visualization of Polysaccharide Degradation by Marine Bacteria. Poster, Ocean Floor Symposium, MARUM.

Kandemiroglu, O.C. & Wisotzky, F. (2018). Vorkommen von organischen Spurenstoffen im Grundwasserleitern in NRW. 26th FH-DGGV Conference, RUB, 21–24.03.2018.

Kandemiroglu, O.C., Loewe, R. & Wisotzky, F. (2012). Hydrogeochemische Untersuchungen im Raum Nettetal zur Klärung der Nickelmobilität. Poster, Bochumer Grundwassertag, RUB.

TEACHING, SERVICE & TRAINING

Teaching — four lecture courses over four semesters, hydrogeochemistry & hydrogeology, RUB (WiSe 2016/17–WiSe 2017/18); RUB Research School Teaching Assistantship (RS PLUS, WiSe 17/18).

Service — Studienkommission FB5, Uni Bremen (2021–2023); elected deputy member, Fachbereichsrat FB5 (2024–2026), independent candidate; Studienlotse (student mentor), Uni Bremen (SoSe 2023–WiSe 2025/26); Campus Guide & school outreach, RUB International Office (2016–2018).

Awards — DAAD STIBET graduation grant (2010) · Ruth und Gert Massenberg Stiftung travel grant (awarded 2017; congress 2016).

LANGUAGES

German — C1, TestDaF 2007 (TDN 4/3/5/4); M.Sc. and doctoral study conducted in German.

- Borehole logging (Full Waveform Sonic, Acoustic Televiwer, Gamma Ray) at 12 boreholes; MATLAB calibration of a thermo-triaxial apparatus; mud sampling per DIN 18123.

Trace-element & aqueous geochemistry — PHREEQC/AquaChem speciation & saturation-index modelling with parallel redox treatments and charge-balance QC · selective extraction (Na-acetate) & aqua regia digestion (DIN EN 13346) · XRF core scanning · AAS & FE-SEM–EDX interpretation.

Computing & bioinformatics — Python (NumPy, SciPy, pandas, matplotlib, seaborn; course 05-MCM-GS-1, Uni Bremen 2023/24–24/25) · three self-written FT-ICR-MS imaging tools (Transformer, Visualizer, Networker) · Galaxy bioinformatics workshop (2025) · Cytoscape · ImageJ · MATLAB/Octave · BAYSPAR, QAnalySeries.

Technical reports & theses

Kandemiroglu, O.C. (2026). Instant Visualization of Organic Matter Transformation in the Ocean. M.Sc. thesis, University of Bremen.

Kandemiroglu, O.C. (2025). Multi-proxy Evidence for Gradual Limnic–Marine Transition in the Black Sea at 8.5–8.0 ka. Technical report, Uni Bremen / MARUM / AWI.

Kandemiroglu, O.C. (2022). Coupled pore-water geochemistry and ¹³C-labelled lipid biomarkers reveal active methane-oxidizing communities across the redox zonation of the Helgoland Mud Area. Technical report, Uni Bremen / MARUM.

Kandemiroglu, O.C. (2011). Untersuchungen zur Herkunft des Nickels im Rohwasser der Wasserwerke Breyell und Kaldenkirchen. M.Sc. thesis, Ruhr-University Bochum.

Full list — ResearchGate profile.

Memberships — EGU (2016) · DAbG (2022) · EANA (2023) · The Geochemical Society (2023) · EAOG (2024).

Schools & workshops — MagellanPlus/IODP³, MARUM (2026) · MarMic "Marine Viruses", MPI-MM (2026) · Visualization in Data Science, Uni Bremen (2025) · Galaxy bioinformatics workshop (2025) · GLOMAR/MARUM Stable Isotopes in Ecology (2022) · GESEP Scientific Drilling (2018) · IGSH Groundwater & Energy (2017) · German Geothermal Congress (2016).

English — C1, UNicert III *Allgemeine Wissenschaftssprache*, grade 1.7 (2020); M.Sc. programme language English.

Turkish — native. References available on request.